

From Control to Offline Reinforcement Learning

Learning Better Decisions from Logged Data

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From Prediction to Decision

Prediction	Decision
What will happen?	What should we do?
Truck arrival time	Which truck to unload next
Grain quality	Where to route the grain
Train delay	Which loading schedule to use

Machine learning describes the operation. Decision-making changes it.

Reinforcement learning is a framework for learning such decisions.

Why Decisions Are Different



- ▶ Supervised learning learns from a fixed dataset.
- ▶ Decision-making selects actions rather than making predictions.
- ▶ Those actions influence future states and therefore the data seen later.

Prediction assumes a mostly stationary data source.
Decisions change the state distribution the model will face next.

How Have We Solved Decision Problems?

**Rules and
expert systems**

Encode operational
knowledge directly.

**Optimization
and control**

Model the system
and compute
good actions.

**Reinforcement
learning**

Improve decisions
using experience
and feedback.

These are not mutually exclusive: practical systems often combine expert knowledge, control, optimization, and learned policies.

The Classical RL Idea



Reinforcement learning learns a decision strategy through interaction.

It can optimize long-term outcomes, account for how one decision affects later decisions, and learn strategies that are hard to specify manually.

But live trial and error is often expensive, slow, disruptive, or unsafe.

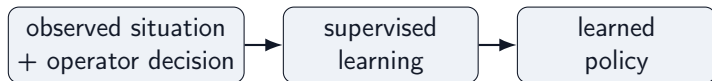
A Modern Starting Point: Learn from Logged Decisions

Classical online RL	Offline approach
act	historical logs
observe	train offline
update	evaluate
act again	cautious deployment

Many real systems already contain years of operational decisions made by people, controllers, or earlier software.

Offline RL asks whether we can learn a better decision strategy from logged interaction data, without new trial and error during training.

Start Simple: Behavior Cloning



- ▶ Observe what an operator or existing controller did.
- ▶ Train a model to predict the same action in similar situations.
- ▶ Deploy the trained model as a learned “policy”.

Behavior cloning is fully offline and requires no simulator or reward model.

Can we do more than learn based on operator data? Can we prefer decisions that led to better long-term outcomes?

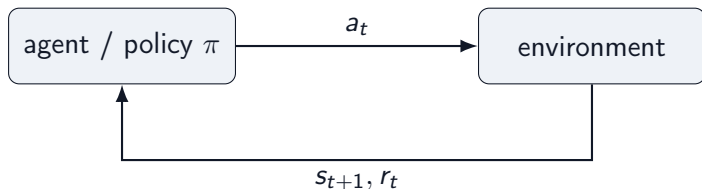
To answer that, we first need a language for decision problems: states, actions, and policies.

States, Actions, and Policies

Object	Grain-terminal example
state s_t	current queue, silo inventory, grain quality, train progress
action a_t	unload truck, choose pit, route conveyor, assign silo, fill car
policy $\pi(a \mid s)$	rule or model that chooses actions from states
next state s_{t+1}	the facility after the action has changed it

A decision problem starts with states, actions, and a policy that maps situations to decisions.

Agent-Environment Loop



- ▶ **Agent:** chooses the next action from the current state.
- ▶ **Environment:** the facility or simulation that returns the next state and reward.
- ▶ **Key contrast:** supervised learning predicts from fixed data; RL actions create the next data.

Imitation Learning: Turn Decisions Into Supervised Learning

First idea: collect observed behavior and train a policy to predict what the operator would do.

Historical demonstrations come from a **behavior policy** π_b , e.g. human terminal operators:

$$\mathcal{D} = \{(s_t, a_t)\}, \quad a_t \sim \pi_b(a \mid s_t)$$

Train π_θ as a supervised action predictor:

Continuous actions

steering angle, conveyor speed

Discrete actions

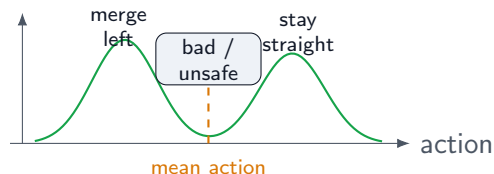
route choice, silo assignment

$$\min_{\theta} \mathbb{E}_{\mathcal{D}} [\|a - \pi_{\theta}(s)\|^2]$$

$$\min_{\theta} -\mathbb{E}_{\mathcal{D}} [\log \pi_{\theta}(a \mid s)]$$

Attractive because it is fully offline, simple to train, and does not require defining a reward.

Pitfall 1: Averaging Demonstrations Can Fail

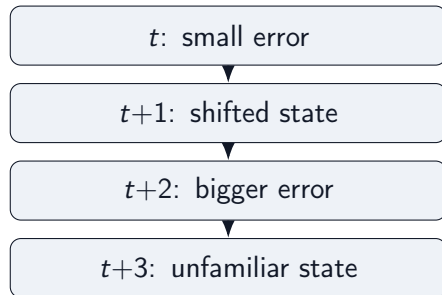


Consider imitation data for self-driving cars:

- ▶ in similar observed states, driver 1 stays in lane
- ▶ driver 2 changes lanes
- ▶ squared-error regression learns the conditional mean action

Problem: the mean action can lie between valid modes: a poor command that no expert demonstrated.

Pitfall 2: Errors Accumulate Over Time



A policy trained only on expert states may not know how to recover from states caused by its own mistakes.

- → Distribution shift: deployment samples from the learned policy's state distribution, not the expert's.

Problem: one-step prediction errors can look small while long rollouts become unstable.

Pitfall 3: Imitation Ignores Outcomes

Behavior cloning asks only:

“What action was taken in this situation?”

It does not ask:

“Which action led to the best long-term outcome?”

Every logged decision becomes a training example. The quality of the outcome does not change how strongly it is learned.

Problem: behavior cloning reproduces past decisions, but it cannot prefer decisions with better long-term consequences.

Rewards Measure the Consequences of Decisions

A reward turns delayed product outcomes into feedback for decisions:

$$r_t = r(s_t, a_t, s_{t+1})$$

For grain loading, rewards can combine:

- ▶ train loading progress and throughput
- ▶ contract quality satisfaction
- ▶ penalties for rehandling, congestion, and delay

Rewards tell the learner what outcomes matter, not just what actions were taken.

Long-Term Consequences

A good decision is one that leads to *high future reward*, not just a high immediate reward.

$$\text{Future Reward} = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$$

where $0 < \gamma < 1$ discounts delayed rewards.

A value function predicts this quantity before making a decision:

$$V(s) \approx \text{future reward, starting from state } s$$

$$Q(s, a) \approx \text{future reward, starting from state } s \text{ and taking action } a$$

Main takeaway

V and Q quantify how good it is to be in a state and to take an action. They are central objects in RL.

Three Common RL Families

Family	Basic idea
Value-based RL	Learn $Q(s, a)$, act greedily
Policy gradient	Directly improve $\pi(a s)$
Actor-critic	Learn both policy and value

- ▶ Q-learning is conceptually simple, but is most natural when actions are discrete.
- ▶ Policy gradients apply more generally, but require expensive rollouts.
- ▶ Actor-critic methods combine both ideas: they learn a policy and a value estimate, and work with discrete or continuous actions.

Learning the Value Function

Learning the Q -function is an important step in nearly all of RL.

To keep things simple, Q satisfies a recursive relation:

long-term value = expected current reward + discounted future value.

$$Q(s_t, a_t) \approx r_t + \gamma V(s_{t+1})$$

This recursion can be solved with exact or approximate dynamic programming.

We will skip those details and use the equation only as the basic intuition behind value learning.

Using Advantage to Weight Data

Advantage scores a logged action relative to what is typical in the same state:

$$A(s, a) = Q(s, a) - V(s)$$

Convert advantage into a training weight:

advantage weight: $w(s, a) = \exp(\beta A(s, a))$, $\beta > 0$

advantage-weighted BC: $\min_{\theta} \sum_{(s_i, a_i) \in \mathcal{D}} w(s_i, a_i) \|\pi_{\theta}(s_i) - a_i\|^2$

- ▶ positive advantage $\rightarrow$ larger weight $\rightarrow$ state/action pair is enhanced in importance
- ▶ negative advantage $\rightarrow$ smaller weight $\rightarrow$ state/action pair is reduced in importance

Simplest offline RL idea

Count “good” data points multiple times, and “bad” data points only as a fraction.

Three Big Ideas in Offline Reinforcement Learning

How have researchers tried to improve behavior cloning?

1. **Reweight the logged data**

AWR: copy successful decisions more strongly.

2. **Learn conservative value estimates**

CQL: be pessimistic where evidence is weak.

3. **Estimate values only from supported actions**

IQL: avoid maximizing over unsupported actions altogether.

Map

The algorithms are examples. The three ideas are the structure of the field.

Problem: Not All Demonstrations Are Equally Useful

Behavior cloning treats every logged decision equally.

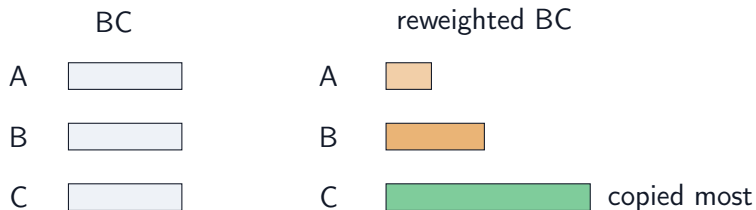
But logged data often contains a mixture:

- ▶ strong operator decisions,
- ▶ reasonable routine decisions,
- ▶ choices that led to delay, rehandling, or poor downstream outcomes.

Question

Can we learn more from the better demonstrations?

Central Idea: Good Demonstrations Count More



Keep supervised learning, but change how much each example matters.

The weight comes from estimated long-term value.

Representative Method: AWR

Advantage-Weighted Regression assigns a weight to each logged decision.

- ▶ estimate long-term value,
- ▶ compare the logged action to typical behavior in the same state,
- ▶ assign a larger weight to positive-advantage decisions,
- ▶ train with weighted supervised learning.

What to remember

Good demonstrations count more.

Representative algorithm: Advantage-Weighted Regression (AWR).

Remaining Problem: Where Do the Weights Come From?

Reweighting is only as good as the value estimates behind the weights.

If the value model is unreliable, the weights become unreliable too.

Next question

How can we estimate values safely from a fixed offline dataset?

Problem: Offline Values Can Be Overconfident

To compute useful weights, we need value estimates.

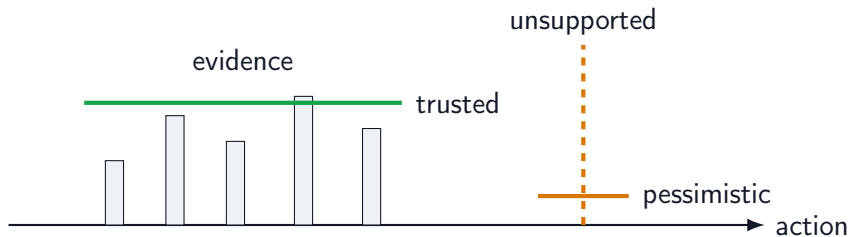
But value estimation is harder offline:

- ▶ online learning can test actions and correct bad guesses,
- ▶ offline learning cannot collect new feedback,
- ▶ unsupported actions may receive overly optimistic predicted values.

Question

How can we learn values without trusting predictions where the data is weak?

Central Idea: Be Pessimistic Where Evidence Is Weak



Conservative value learning lowers confidence away from the logged data.

The policy should not be attracted to actions whose value is high only because the model is guessing.

Representative Method: CQL

Conservative Q-Learning changes the value-learning objective.

Its principle is simple:

- ▶ fit observed transitions in the dataset,
- ▶ penalize high values assigned to actions not well supported by the dataset,
- ▶ prefer policies whose improvement is backed by evidence.

What to remember

Do not trust unsupported actions.

Representative algorithm: Conservative Q-Learning (CQL).

Remaining Problem: Can We Avoid the Risk Entirely?

CQL makes unsupported actions less attractive.

But it still starts from the same danger:

searching over actions can expose value errors.

Next question

Can we estimate values without maximizing over unsupported actions at all?

Problem: Can We Avoid Unsupported Maximization?

Instead of correcting bad value estimates, can we avoid creating them?

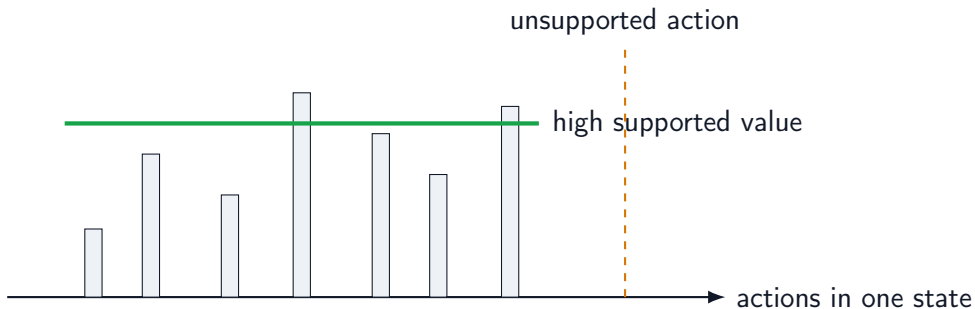
The risky question is:

What action has the highest predicted value?

IQL asks a safer question:

What good actions are already supported by the data?

Central Idea: Learn Values Inside the Dataset



IQQL estimates high values among logged actions, not over all possible actions.

In practice, this is implemented with expectile regression: a soft way to estimate a high supported value.

Representative Method: IQL

Implicit Q-Learning separates value estimation from action maximization.

- ▶ learn values from logged state-action pairs,
- ▶ avoid explicit maximization over unsupported actions,
- ▶ compute advantages using these in-sample values,
- ▶ train the policy with advantage-weighted behavior cloning.

What to remember

IQL = better value estimation + the same policy update as AWR.

Remaining Problem: What Should the Policy Look Like?

The first three ideas focus on value estimation.

But policy representation can also matter.

If several distinct actions are good, a simple policy may average them into a poor action.

Optional next question

How should we represent policies when several actions are equally good?

Complementary Issue: Several Good Actions



The first three ideas focused on value estimation.

IDQL addresses a different issue: representing policies when several actions are equally good.

Diffusion Policies Represent Richer Action Distributions



A diffusion policy can sample several plausible actions for the same state.

IDQL combines IQL-style value learning with diffusion-based policy extraction.

Summary: Three Ideas, Three Representatives

Problem	Core idea	Representative algorithm
Not all demonstrations are equally useful	Reweight data	AWR
Values are unreliable outside the data	Conservative value learning	CQL
Avoid unsupported maximization	In-sample value learning	IQL

Main message

Modern Offline RL is largely about obtaining reliable value estimates from fixed logged data.

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